PHYS 7501:
Particles and Nuclei I

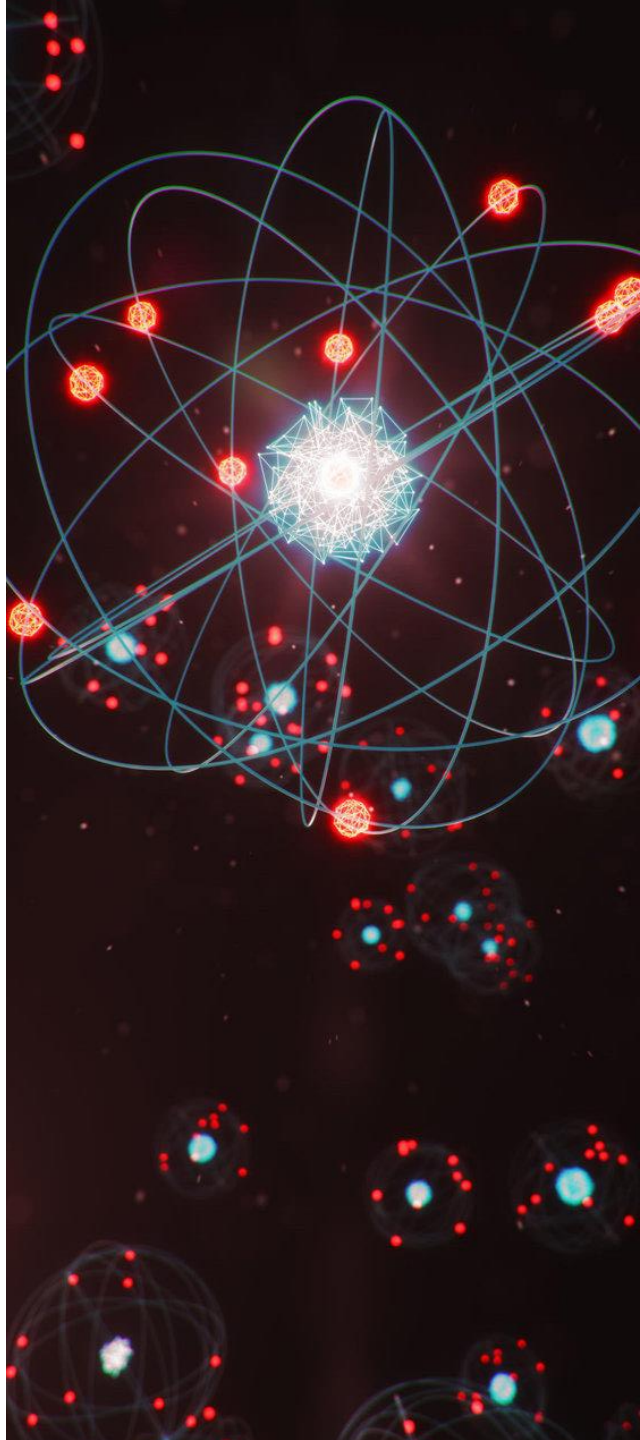
Tues./Thurs. 2 – 3:20PM
Clippinger 105

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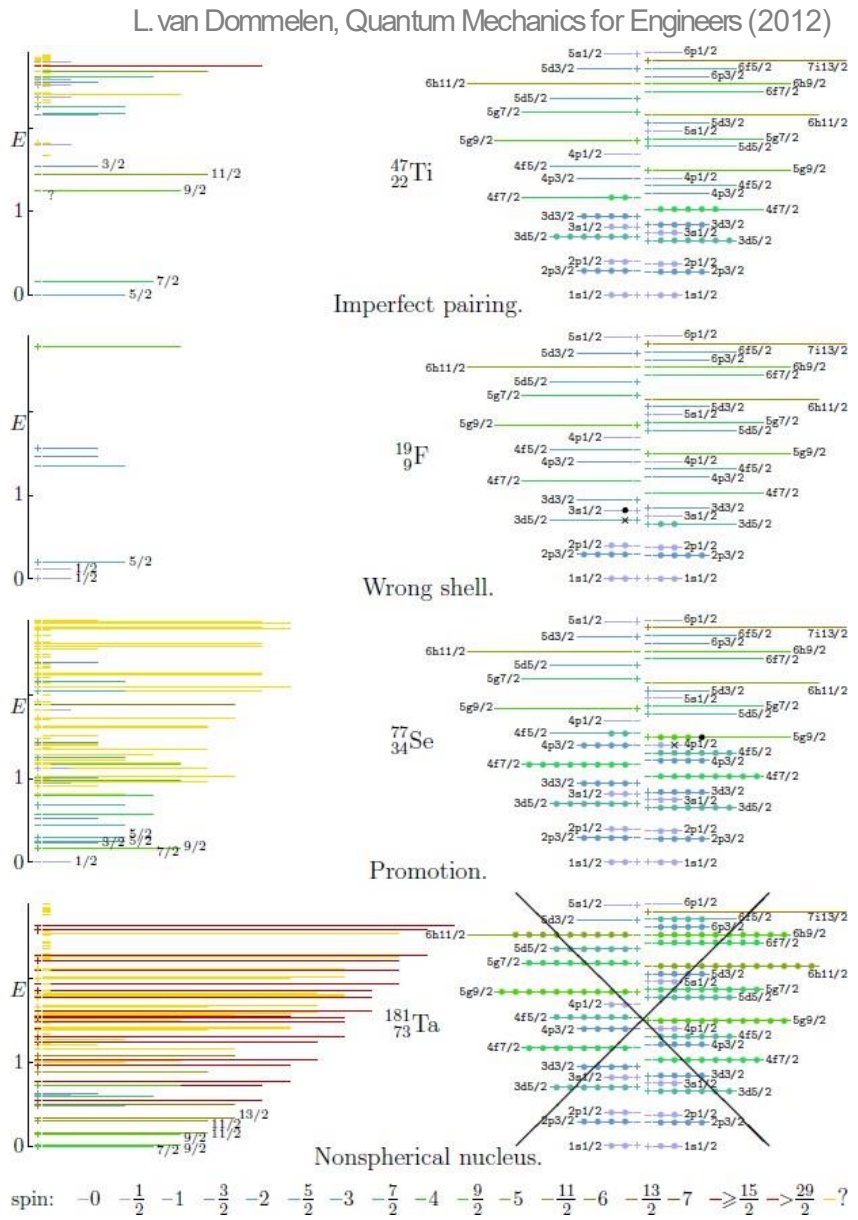
NUCLEAR STRUCTURE

Modern Nuclear Chemistry, Chapter 6

Monday	Tuesday	Wednesday	Thursday	Friday
9/15	9/16	9/17	9/18	9/19
	Nuclear Structure		**Radioactive Decay**	
9/22	9/23	9/24	9/25	9/26
	Alpha Decay HW #2 Due		**Beta Decay**	
9/29	9/30	10/1	10/2	10/3
	Gamma Decay		**Fission**	

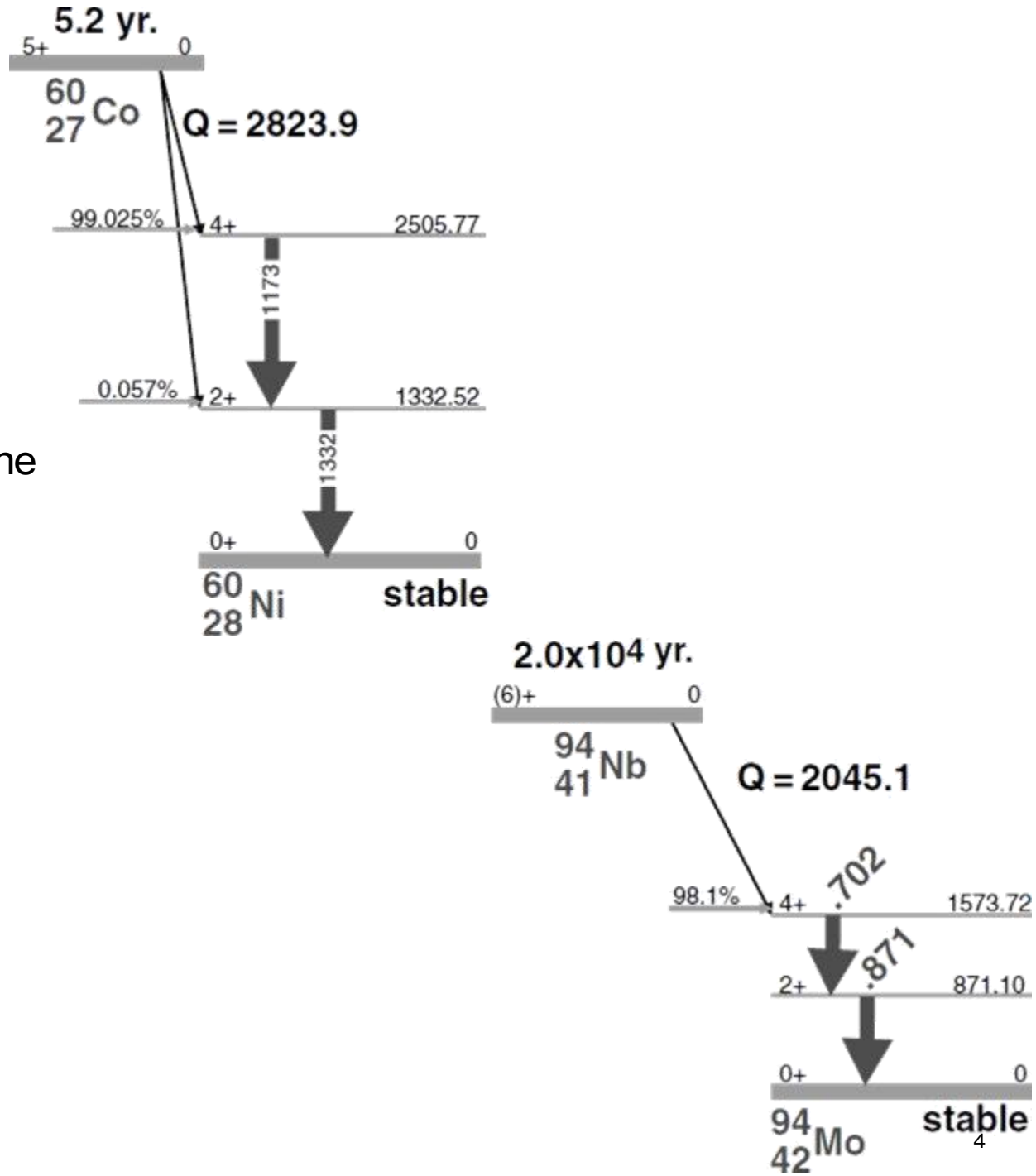
Limitations of the Shell Model

- The Shell Model often isn't as good as you'd hoped....
- Various reasons for the places it fails, however most times, premise of the calculation is incorrect
- Shell Model calculations (generally) assume
  - Minimally interaction nucleons (mostly independent particles, except for pairing)
  - Spherical inert cores of nucleons
- What's the solution?
  - Modify the Shell-Model calculation to be more realistic
  - Use a different model



Shell Model Limitations

- For example, nuclei have series of states that are spaced in energy and linked via transitions that can be described by a collective rotation/vibration of the nucleus.
- The rotational bands of even-even nuclei link the ground state to $2^+,4^+,6^+,8^+$, etc., excited states
- Another example is the inability to predict nuclear masses.
- Shell model potentials must be adjusted to reproduce the ground-state binding energy
- Some other approach, such as a collective model (e.g. the liquid drop model) is needed instead



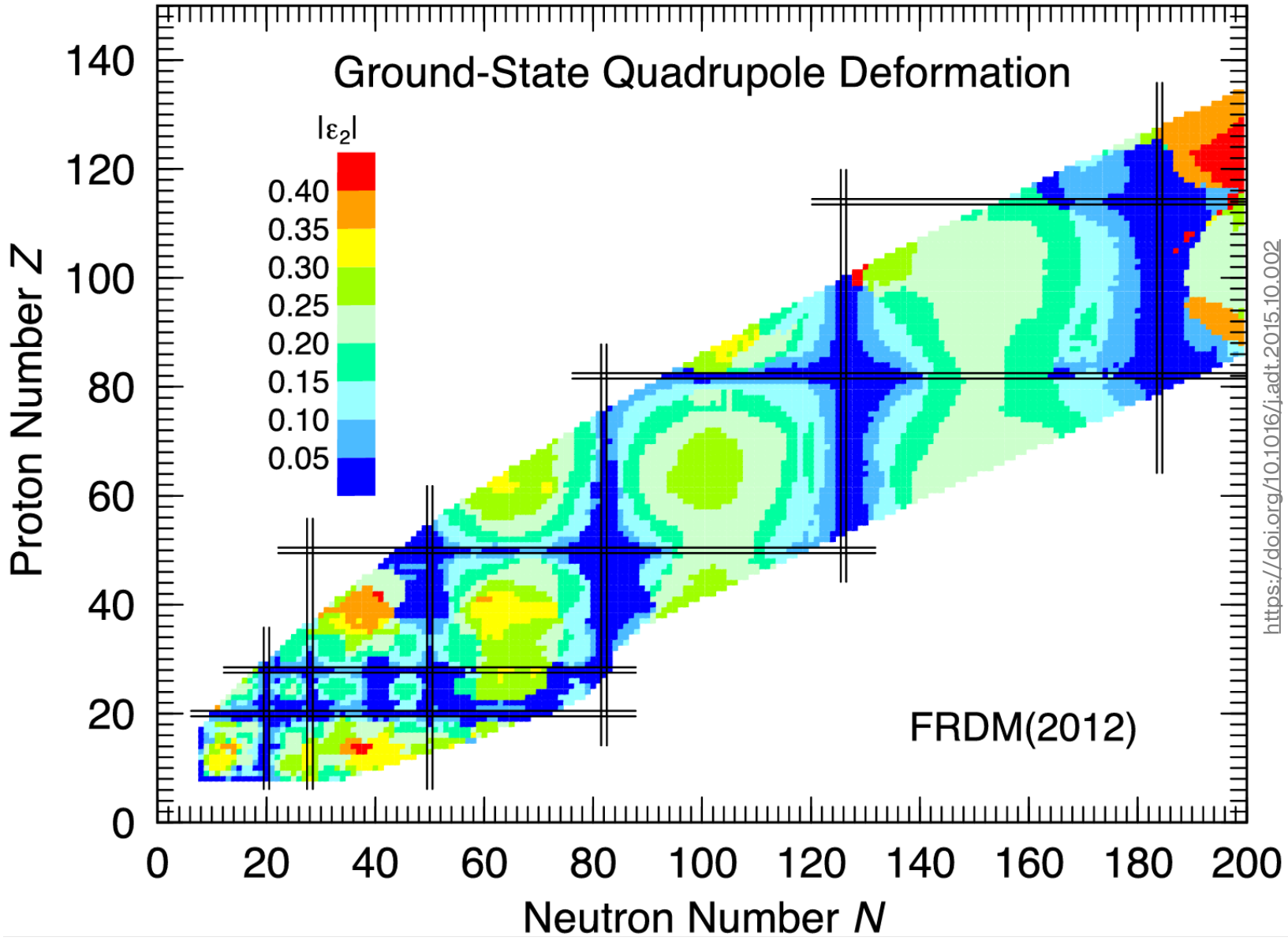
Nuclear Models

- No useful fundamental & universal models exist for nuclei
  - Based on the nuclear interaction, how do we describe all nuclear properties?
  - Promising approaches include “ab initio” methods such as Greens Function Monte Carlo, No-Core Shell Model, Coupled Cluster Model, and Density Functional Theories
- Generally, one of two classes of models are used instead
  - **Independent particle models**
    - A nucleon exists within a mean field (maybe having a few interactions)
    - Examples: The Shell Model, The Fermi Gas Model
  - **Collective Models**
    - Groups of nucleons act together (maybe involving Shell Model aspects)
    - Examples: Liquid Drop Model, Collective Model

Collective Model

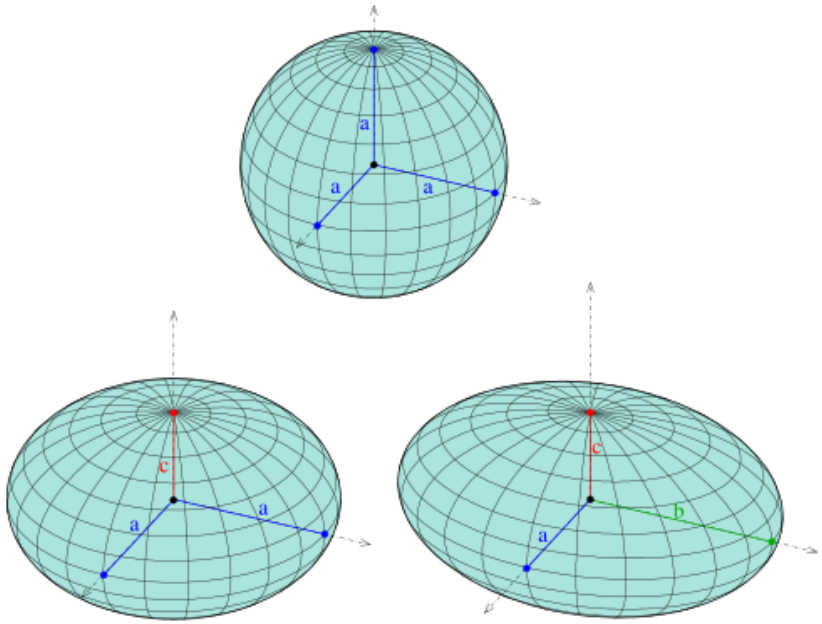
- Nucleons residing in well-defined orbitals in the nucleus can be understood with *relatively* simple math using the Shell Model
  - Properties of the nucleus are dominated by one or two unpaired nucleons
- However, the bulk of the nucleons, which may even be hundreds, only contribute to the central potential and not the nuclear properties
- In reality, we cannot ignore the core nucleons because they give rise to large-scale macroscopic behavior that can be different from the behavior of single particles
  - Independent vs. collective motion
- Two important collective motions:
  - Rotation of deformed nuclei
  - Vibrations of the nuclear shape

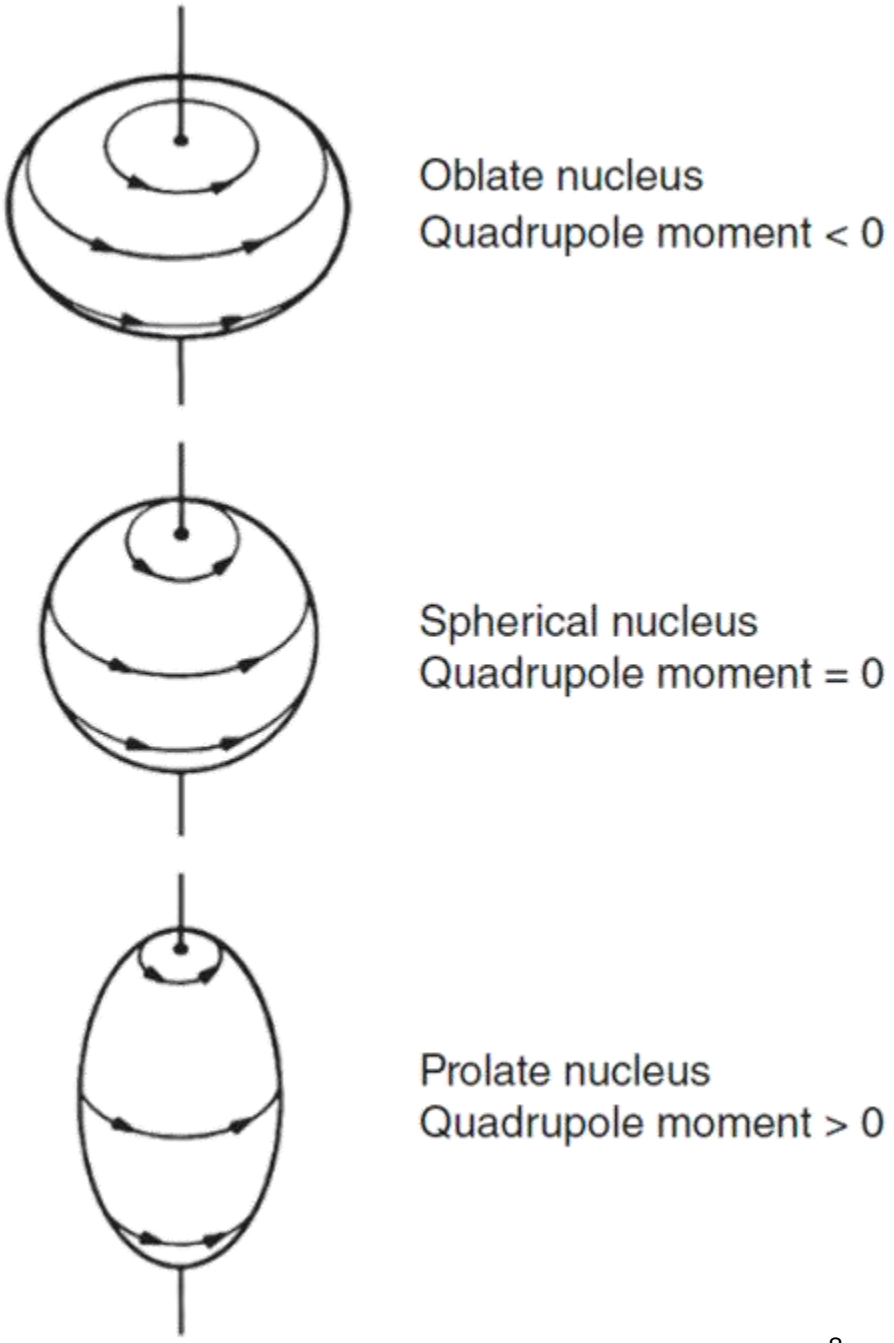
Predicted Regions of Deformation



Nuclear Shapes Reminder

- Treat the nucleus as an ellipsoid (3D object formed by rotation of an ellipse around one of its two major axes)
- **Prolate deformation:** one axis is longer relative to the other two (rugby ball ”rug ball”)
- **Oblate deformation**: one axis is shorter than the other two (pancake)

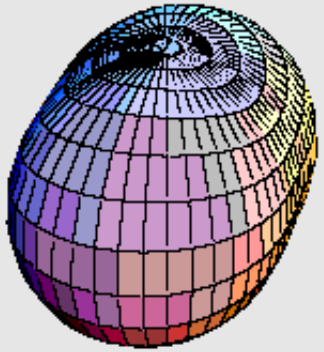




Collective Motion

- Rotational and vibrational motion have unique signatures in excited states
- Relative energies of rotation vs. vibration can be inferred from their geometries
  - Rotational frequency should be similar to $\omega_r \propto \frac{1}{R^2}$ because the moment of inertia is $I = \frac{L}{\omega}$ and $I \propto MR^2$
  - Vibrational frequency should be $\omega_v \propto \frac{1}{(\Delta R)^2}$ because it's like an oscillator
  - $\omega_r \ll \omega_v$

Rotational Motion



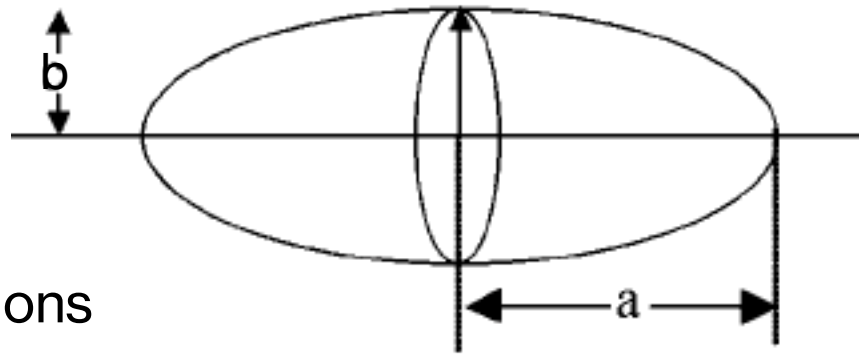
https://web2.ph.utexas.edu/~coker2/index.files/rotation2.htm

Vibrational Motion



https://web2.ph.utexas.edu/~coker2/index.files/rotational.htm

Rotational Motion



- Characteristic of non-spherical nuclei
- Deformation can be permanent or induced by reactions
- Challenging to describe in a single particle framework but the basic behavior can be described using a good parameterization of the shape of the nucleus
- Consider an ellipsoid with semi-major and semi-minor axes, *b* and *a*.
- Surface of the ellipsoid can be written using the expansion

$$\boldsymbol{R(\theta,\phi) = R_{average}[1 + \beta Y_{20}(\theta,\phi)]}$$

$R_{average}$ is the average radius of the two major axes

$$R^2_{average} = \frac{a^2 + b^2}{2}$$

β is a dimensionless measure of deformation

$Y_{20}(\theta,\phi)$ is the spherical harmonic function

- The deformation parameter can be written as

$$\boldsymbol{\beta = \frac{4}{3}\sqrt{\frac{\pi}{5}}\frac{b-a}{R_{average}}}$$

$\boldsymbol{\beta > 0}$ Prolate Shape

$\boldsymbol{\beta < 0}$ Oblate Shape

Rotational Motion: Rigid Rotor

- Energy associated with a rotating object is: $E_{rot} = \frac{1}{2} I\omega^2$
- But, we need this to reflect a quantum mechanical system where $J = I\omega$
- So, $E_{rot} = \frac{1}{2}\frac{J^2}{I}$ and J is quantized making the rotational energy:

$$E_{rot} = \frac{j(j+1)\hbar^2}{2I}$$

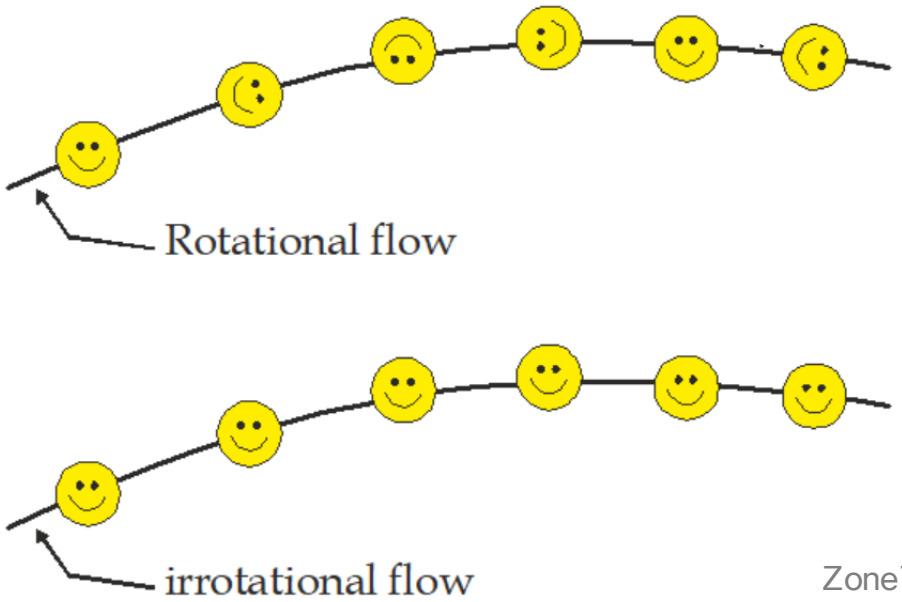
- The rotating nucleus will have excited states spaced by $j(j+1)$ corresponding to rotation
- To calculate moment of inertia, you can consult formulae in the Bohr & Mottelson textbook
  - Generally, for a solid constant-density ellipsoid,

$$I_{rigid} = \frac{2}{5} MR^2\left(1 + 0.31\beta + 0.44\beta^2 + \ \ldots\right)$$

Rotational Motion: Irrotational Motion

- Rather than the whole nucleus rotating, a tide-like effect can produce something like rotation
- Here nucleons move in and out in a synchronized fashion (like people doing the wave)
- Nucleons aren't orbiting, but are bobbing in and out and this motion is called *irrotational*
- There are some solutions for the moment of inertia for a continuous classical fluid

$$I_{irro} = \frac{9}{8}\pi M R^2 \beta^2$$



ZoneTech

Empirical Moment of Inertia

- $E_{rot} = \frac{j(j+1)\hbar^2}{2I}$, so measuring the ΔE between levels will yield the moment of inertia I
- Generally, $I_{irro} < I_{expt} < I_{rigid}$

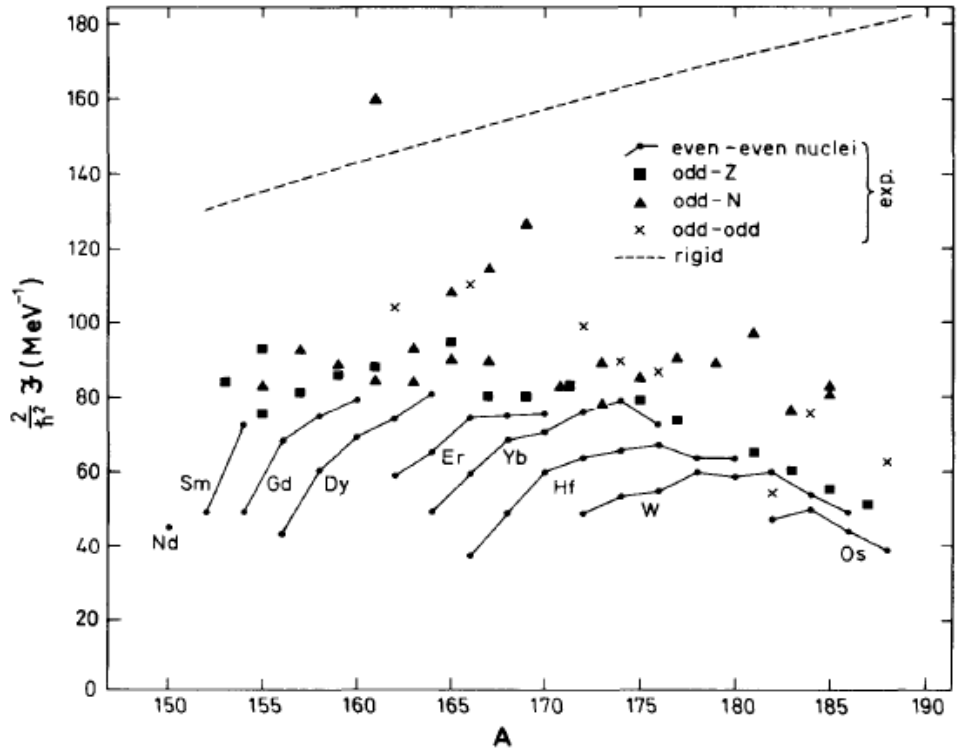


Figure 4-12 Systematics of moments of inertia for nuclei with $150 \leqslant A \leqslant 188$. The moments of inertia are obtained from the empirical energy levels in *Table of Isotopes* by Lederer *et al.*, 1967.

Bohr & Mottelson, Nuclear Structure Vol. II

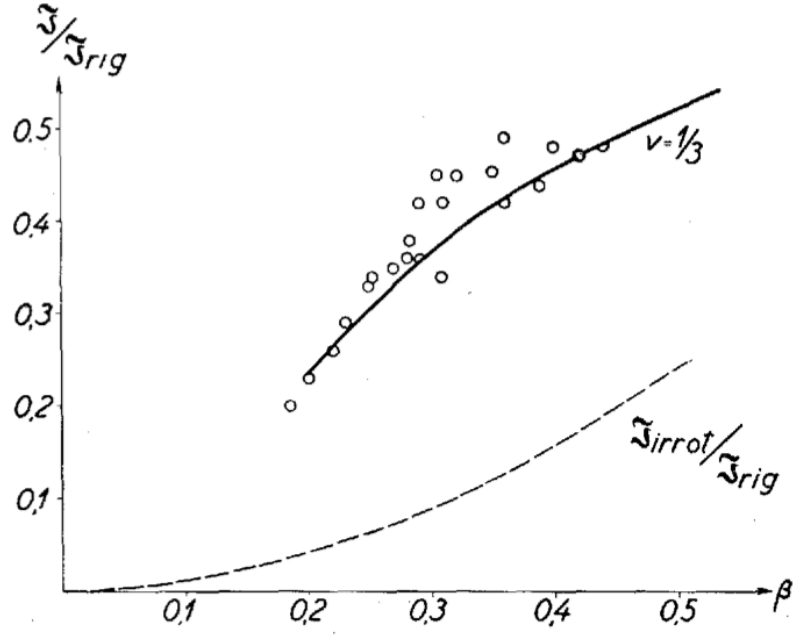
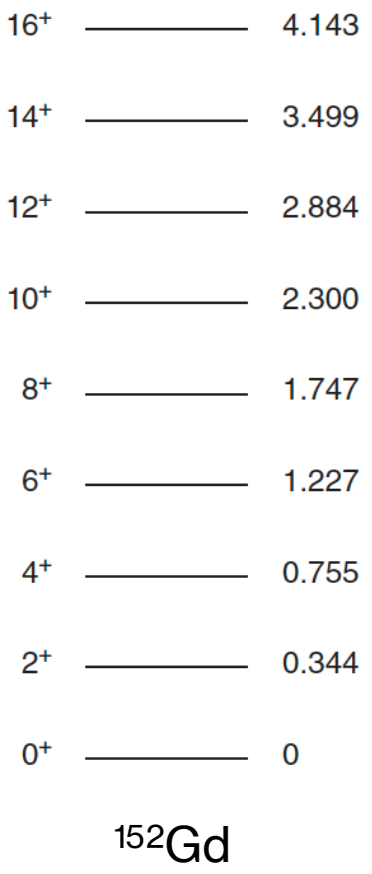


Fig. 2. Dependence of Nuclear Moments of Inertia on the Nuclear Deformation. The empirical moments of inertia for even-even nuclei in the region $150 \leqslant A \leqslant 188$ are plotted as a function of the nuclear deformation. The moments of inertia, obtained from the data in Table I, are given in units of the rigid moment (18), while the deformation parameters β are obtained from the Q_0-values in Table I by means of (19). The nuclear radius has been taken to be $R_0 = 1.2\, A^{1/3}\, 10^{-13}$ cm. The full-drawn curve represents a theoretical estimate, based on the two-nucleon model with an interaction parameter $v = 1/3$ (cf. Fig. 1). For comparison, the moment of inertia corresponding to irrotational flow is shown by the

Bohr & Mottelson Matematisk-fysiske Meddelelser, **30**, (1955)

Rotational Bands

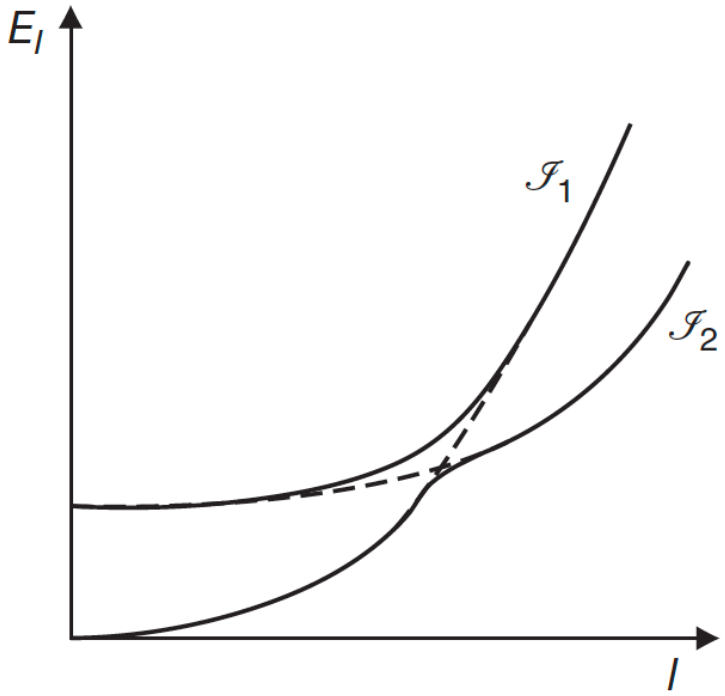
- Sequences of excited states
- $\boldsymbol{E_{rot} = \frac{j(j+1)\hbar^2}{2I}}$, so for a given I, $\Delta \boldsymbol{E_{rot} \propto j(j+1)}$
- Note that parity must be maintained because rotation is symmetric upon reflection. 0^+ ground states can only have $\boldsymbol{j = 0^+, 2^+, 4^+}$, ...
- Note that this interpretation works for even-even nuclei. Odd nuclei are another deal....
- Rotational bands would be challenging to observe without measuring a full level scheme with coincidences

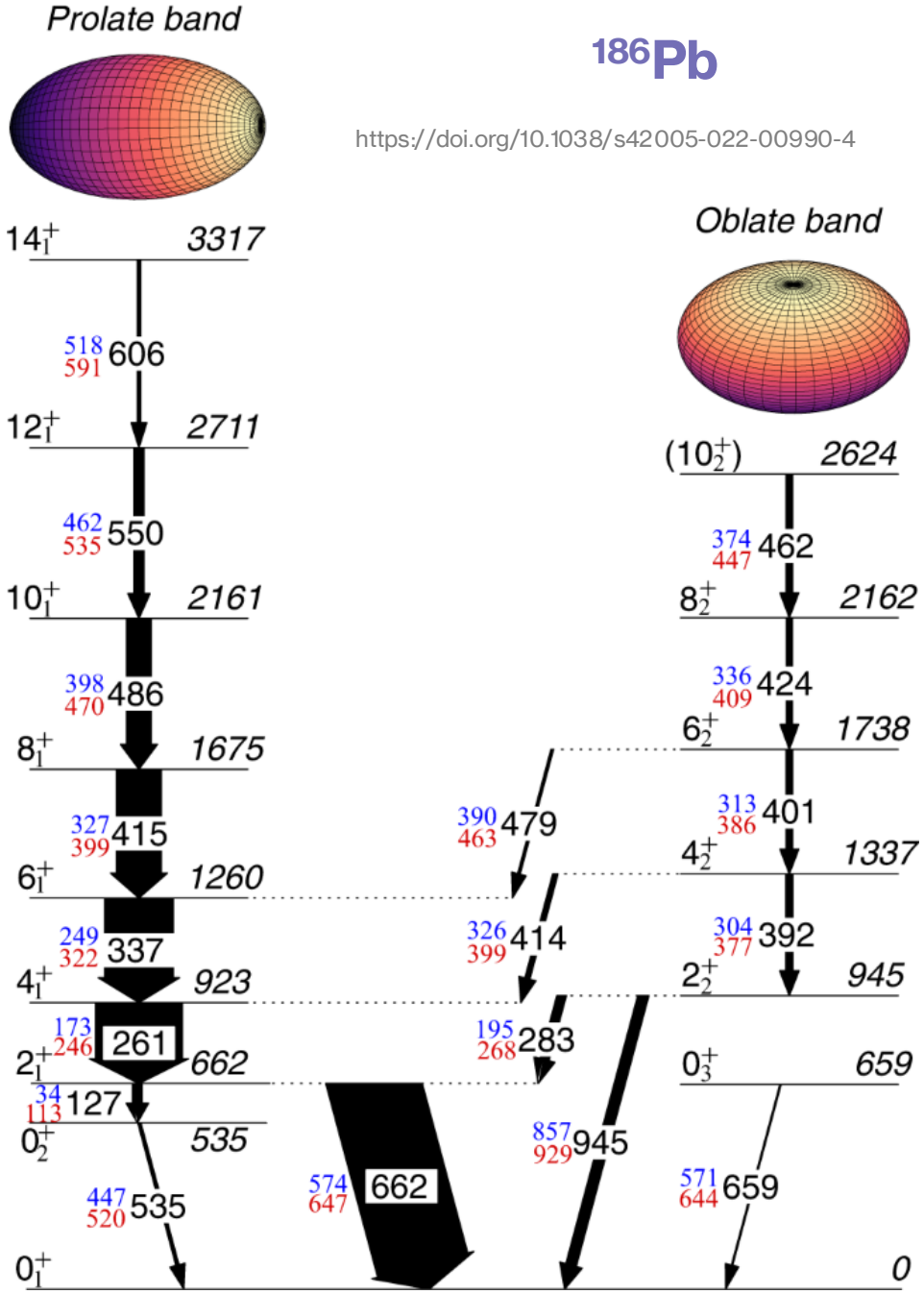


^{152}Gd

Rotational Bands

- Rotation can exist on top of other excitations
- Nuclei can have several rotation bands, and the moment of inertia is often different per band
- Different moments of inertia lead to different energy spacings for the bands

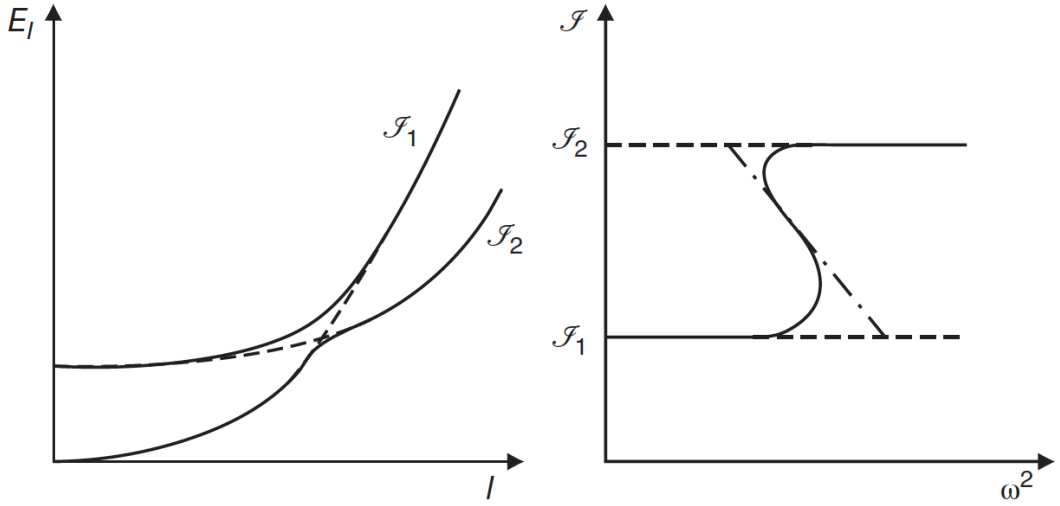


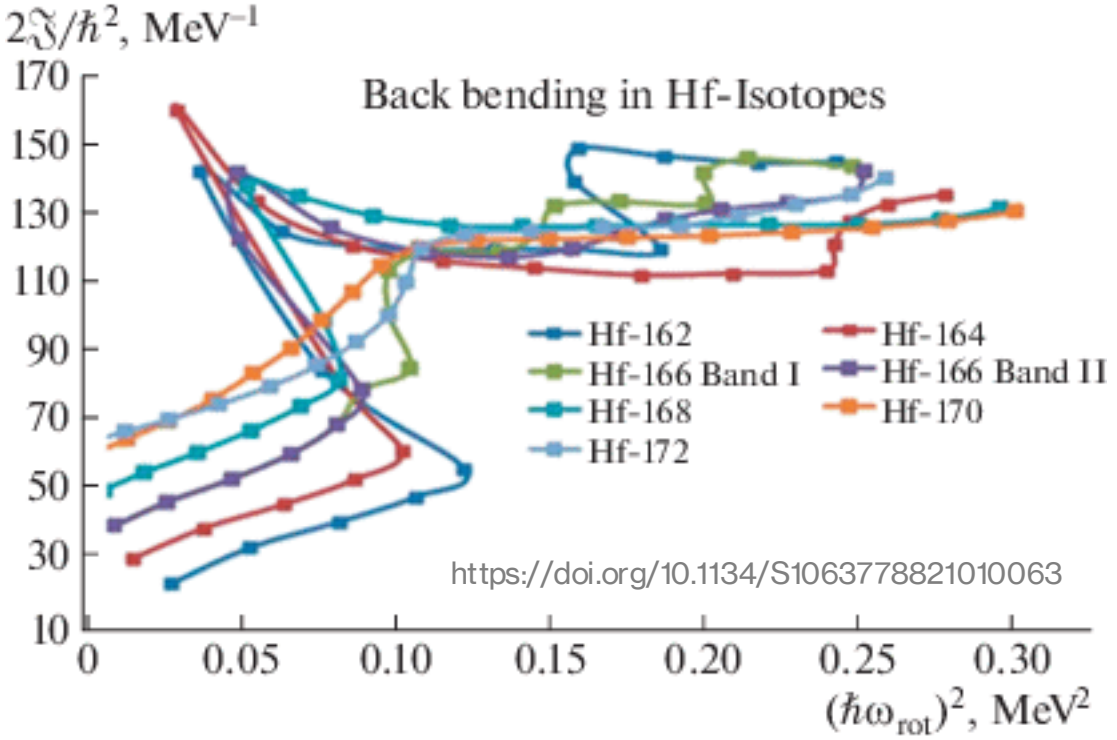


Rotational Bands

- The different I for different rotational bands manifests the so-called “backbend”

- This is when we follow the lowest-energy state for a given spin-parity (called the “yrast” state) belonging to a given rotation band and plot the moment of inertia and the square of the rotational frequency

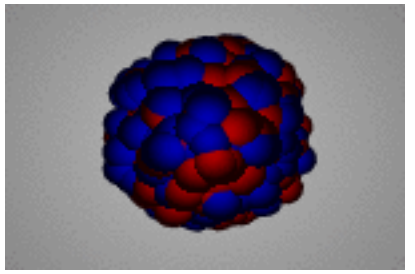




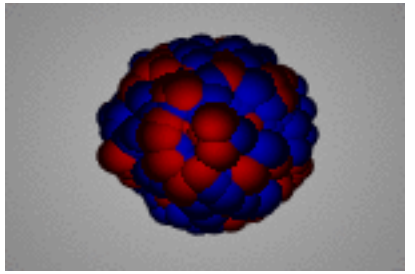
Vibrational Motion

- Considering the nucleus as a liquid drop, the nuclear volume should be able to vibrate
- Imagine the surface of the nucleus vibrating harmonically
  - We can parameterize the surface oscillations in terms of the spherical harmonic functions
  - We label these quanta (phonons) as λ
- Several multipoles are possible
  - Monopole: in & out motion ($\lambda = 0$)
    - "breathing mode"
  - Dipole: sloshing back & forth ($\lambda = 1$)
  - Quadrupole: alternating compression & stretching ($\lambda = 2$)
  - Octupole: alternating pinching on one end and then the other ($\lambda = 3$)
- Not all nucleons need to move together

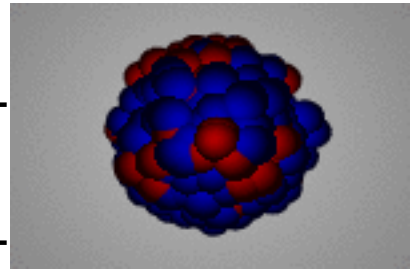
monopole



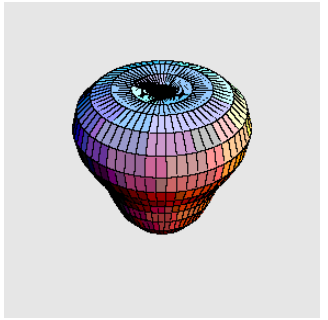
dipole



quadrupole



octupole



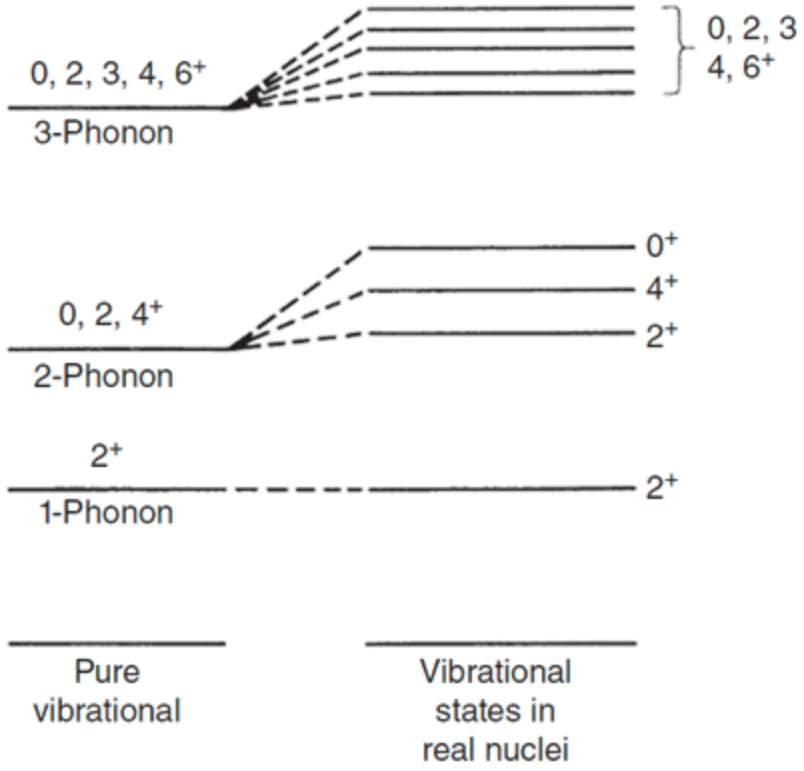
https://web-docs.gsi.de/%7Ewolle/TELEKOLLEG/KERN/index-s.html

- Recall that the energy levels of a quantum mechanical harmonic oscillator are all equally spaced having energies

  $E_N = (N + \frac{1}{2})\hbar\omega_0,\ N = 0, 1, 2, \ldots$

- There is some oscillating deviation from the default shape and a restoring force attempts to return the situation to the default shape

- Just as the quantum harmonic oscillator eigenvalues are quantized, so too will the energy levels for different quanta (phonons) of a vibrational mode



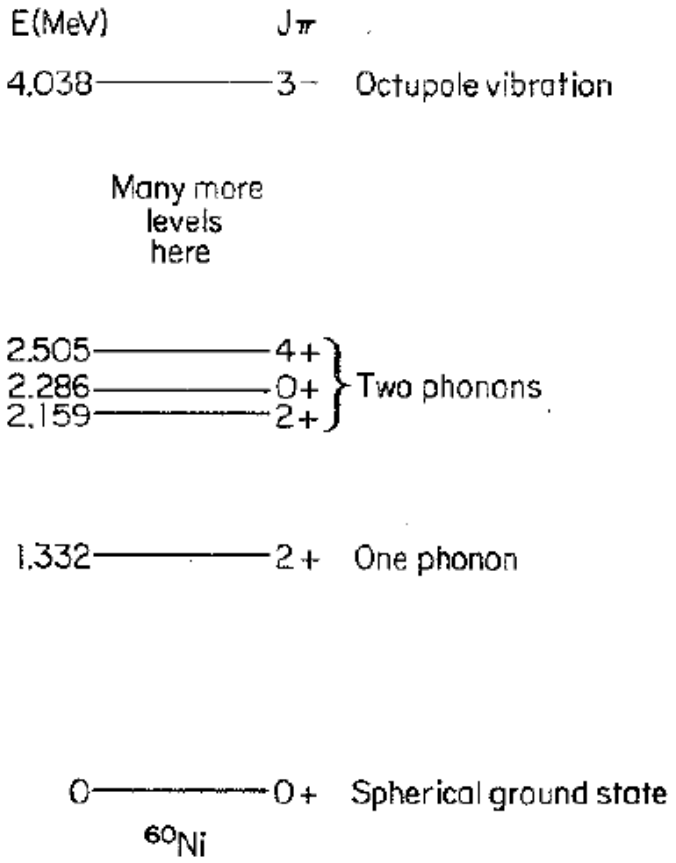
- Even-even nuclides have 0^+ ground states and thus for a $\lambda = 2$ vibration, $\mathrm{N} = 2$ excitations will maintain the symmetry of the wave function. ($\mathrm{N} = 1$ would violate parity)

- Therefore, the first vibrational state will be a 2^+

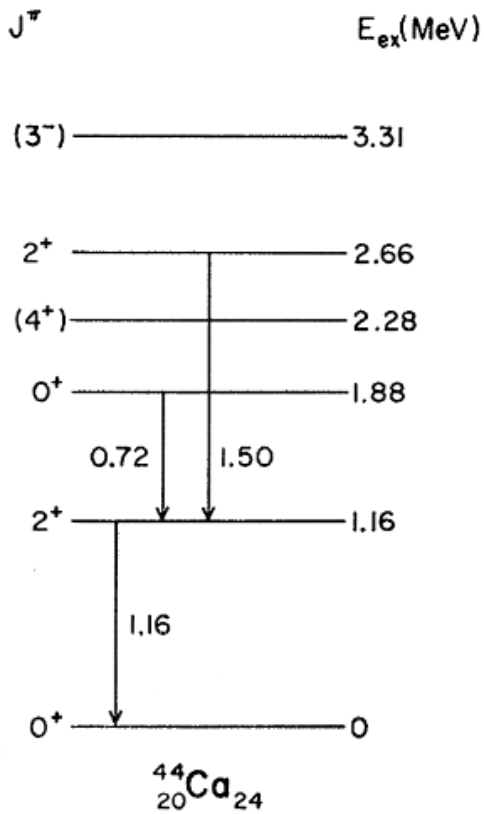
- We can excite an independent quadrupole vibration by adding a second phonon

- The second phonon will build excitations on the first, coupling to either $0^+, 2^+$, or 4^+

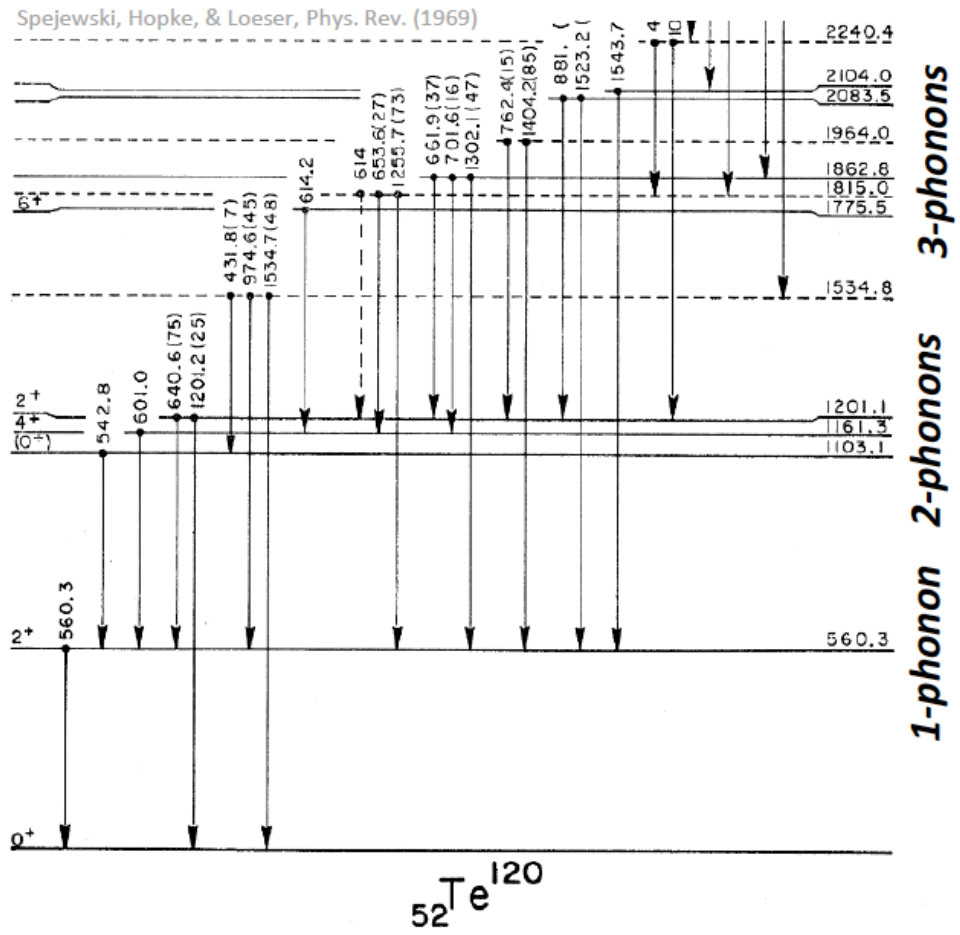
Vibrational Energy Levels



G. Harvey, Introduction to Nuclear Physics and Chemistry (1962)



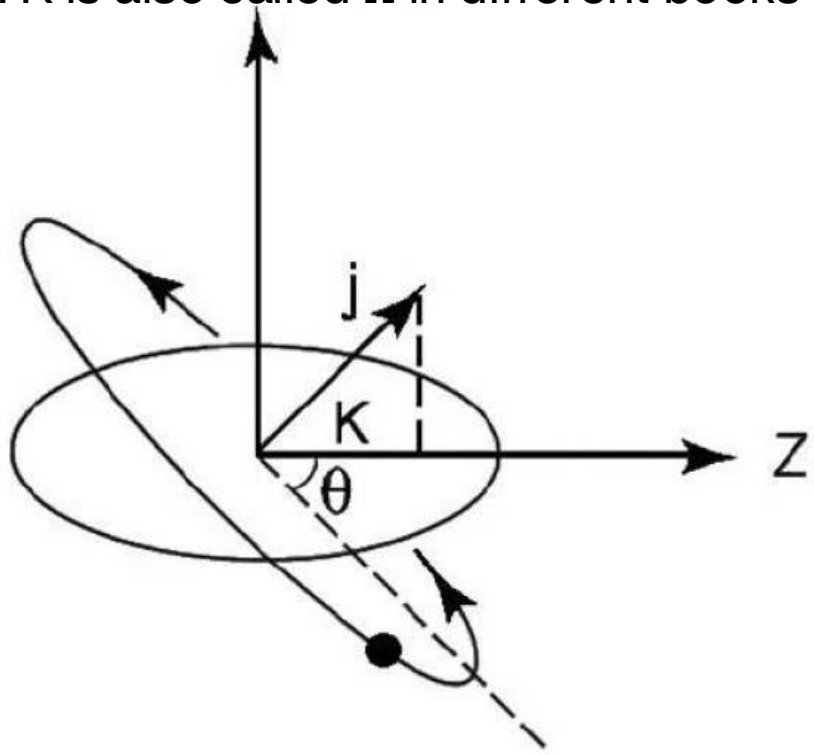
Matin, Church, & Mitchell Phys. Rev. (1966)



Typical signature for vibrating (spherical) nuclei is that $\frac{E(4_1^+)}{E(2_1^+)} \approx 2$ …. This won't be the case for deformed nuclei

Nilsson Model

Note: K is also called Ω in different books

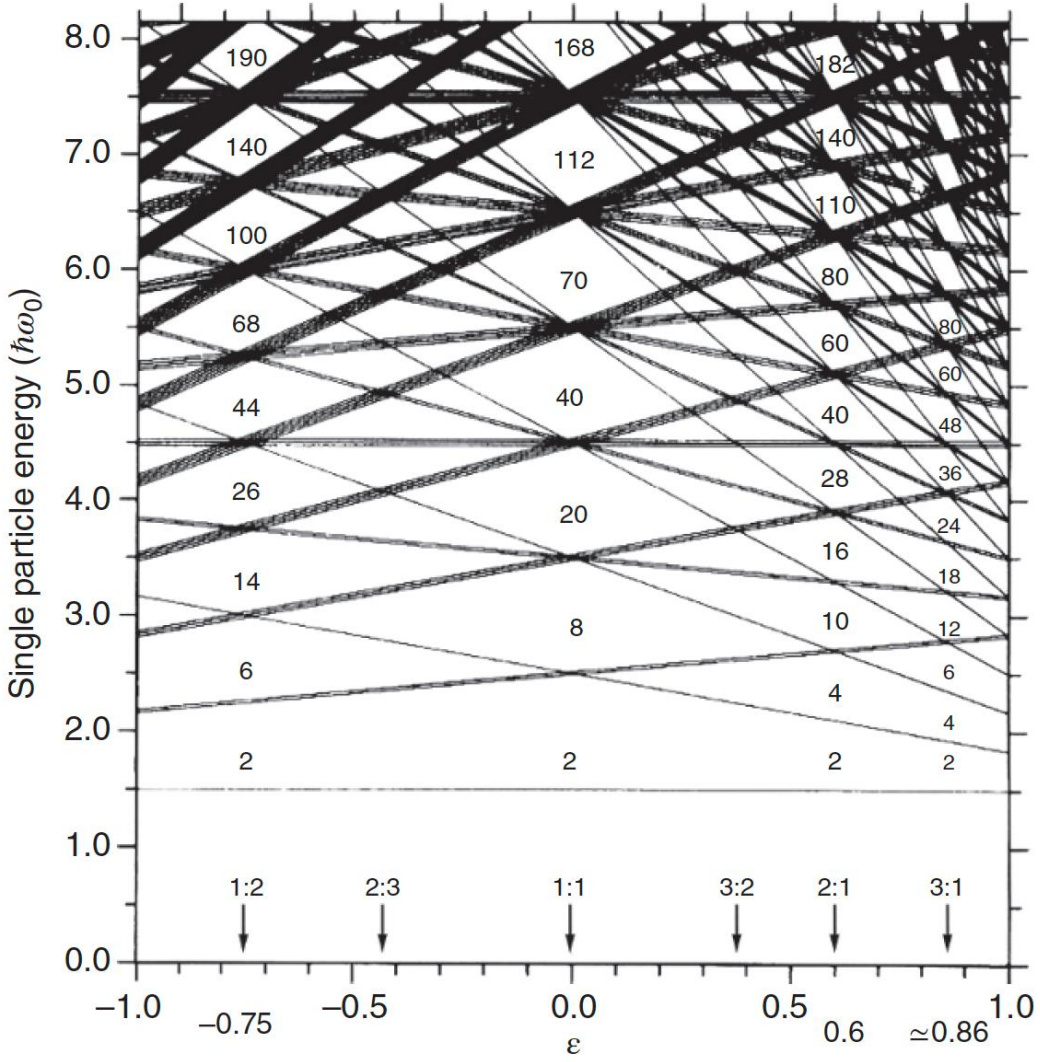


R.Casten, Nuclear Structure from a Simple Perspective (1990)

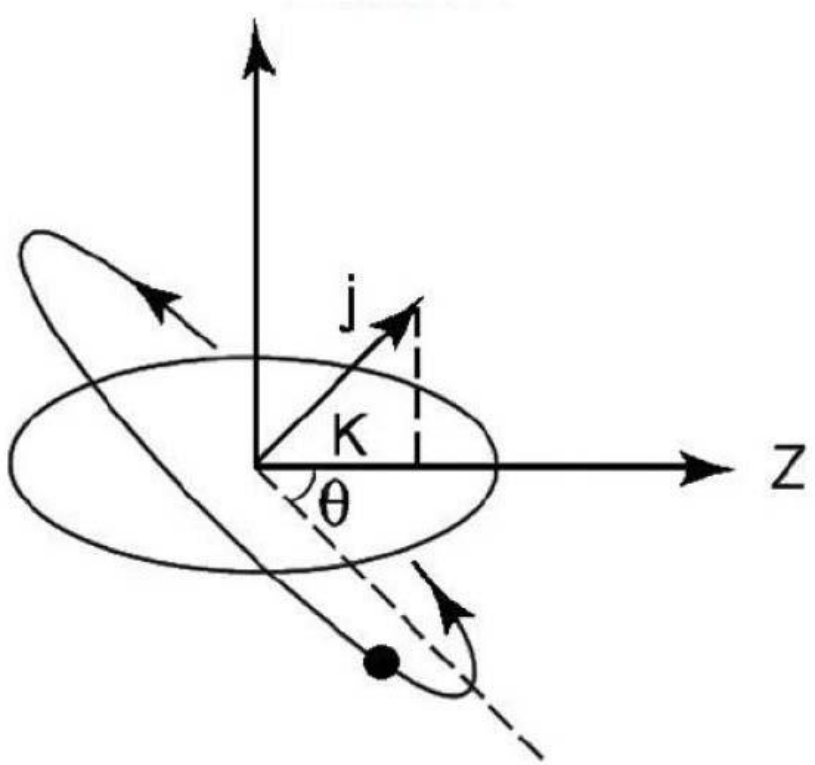
- Until now, we've discussed the two extremes of nuclear structure: **independent particle motion** and **collective motion**
- We can't use our traditional shell model (that we know and love) due to deformation
- Consider a deformed nucleus with axial symmetry that has a single unpaired nucleon orbiting the nucleus
  - We will only consider axially symmetric deformation in this class for simplicity
- The nucleon has some spin, j, which has a projection onto the symmetry axis, K
- j has $\frac{2j+1}{2}$ possible projections of K using a deformed Harmonic Oscillator potential
- Therefore, each single-particle state from our Shell Model splits into multiple states, identified by their K, each of which can contain two particles (spin up and down) for that projection

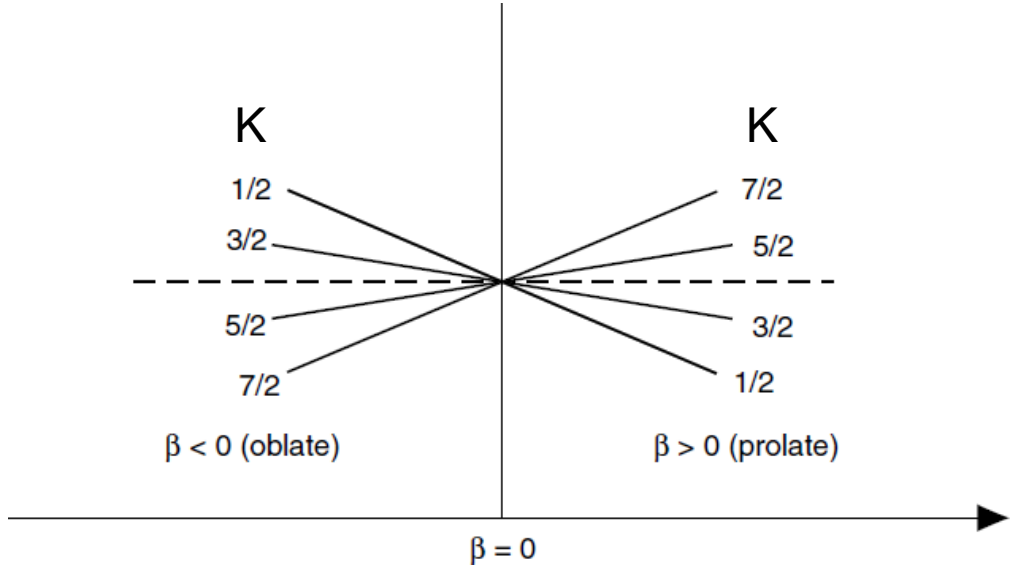
Nilsson Model: Single-Particle Levels

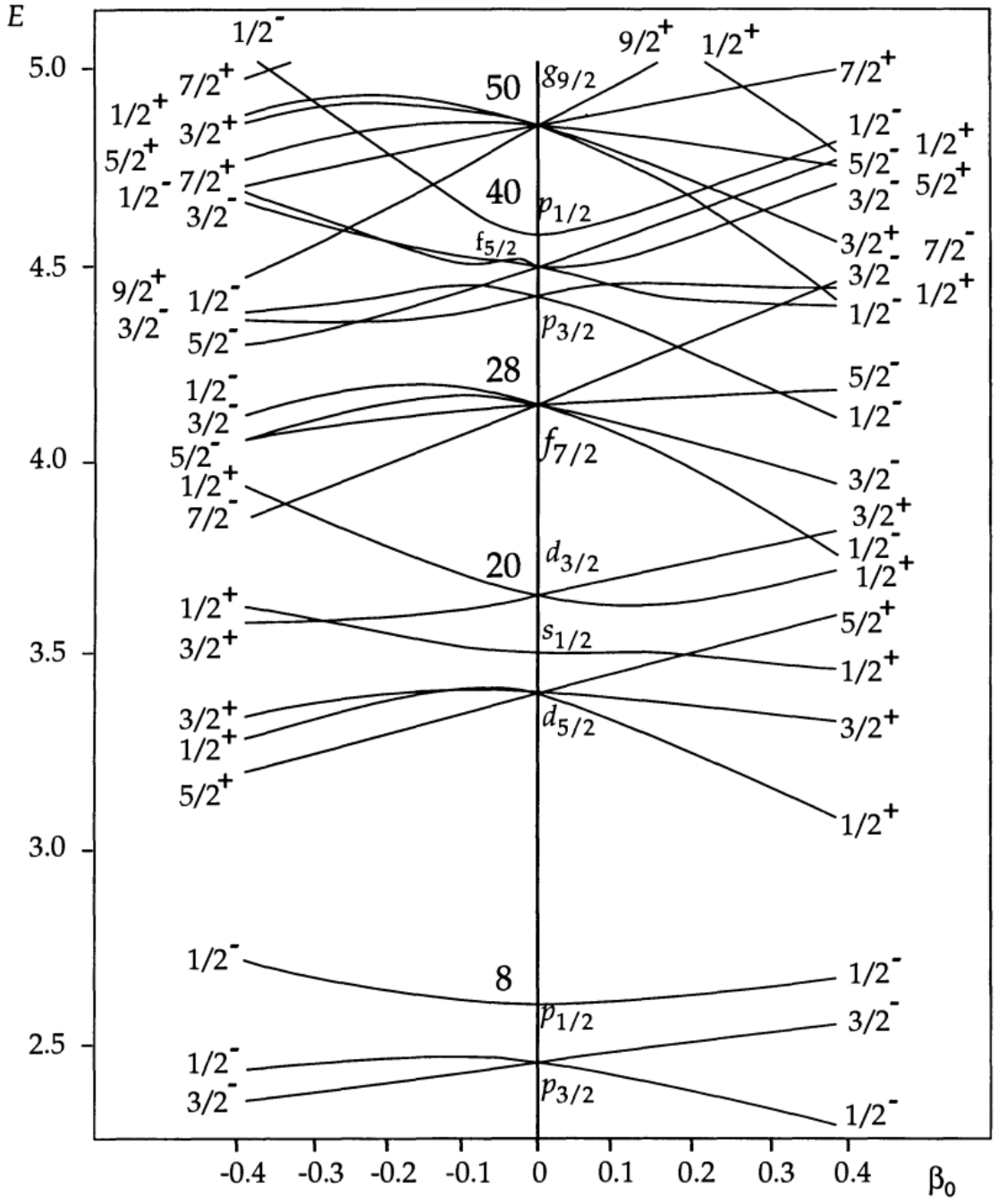
- In the spherical case, the energy levels of each single particle state have a degeneracy of $(2j+1)$
- If the potential has a deformed shape, this will no longer be true... the energy levels in the deformed potential depend on the spatial orientation of the orbit
- So, the energy depends on the component of j along the symmetry axis
- ϵ and β are both deformation parameters that can be used interchangeably
- $\epsilon = \beta = 0$ is spherical, so we see our normal HO levels
- As deformation changes, levels move in energy and new shell gaps (magic numbers) appear when the ratio of semi-major to semi-minor axes are whole numbers

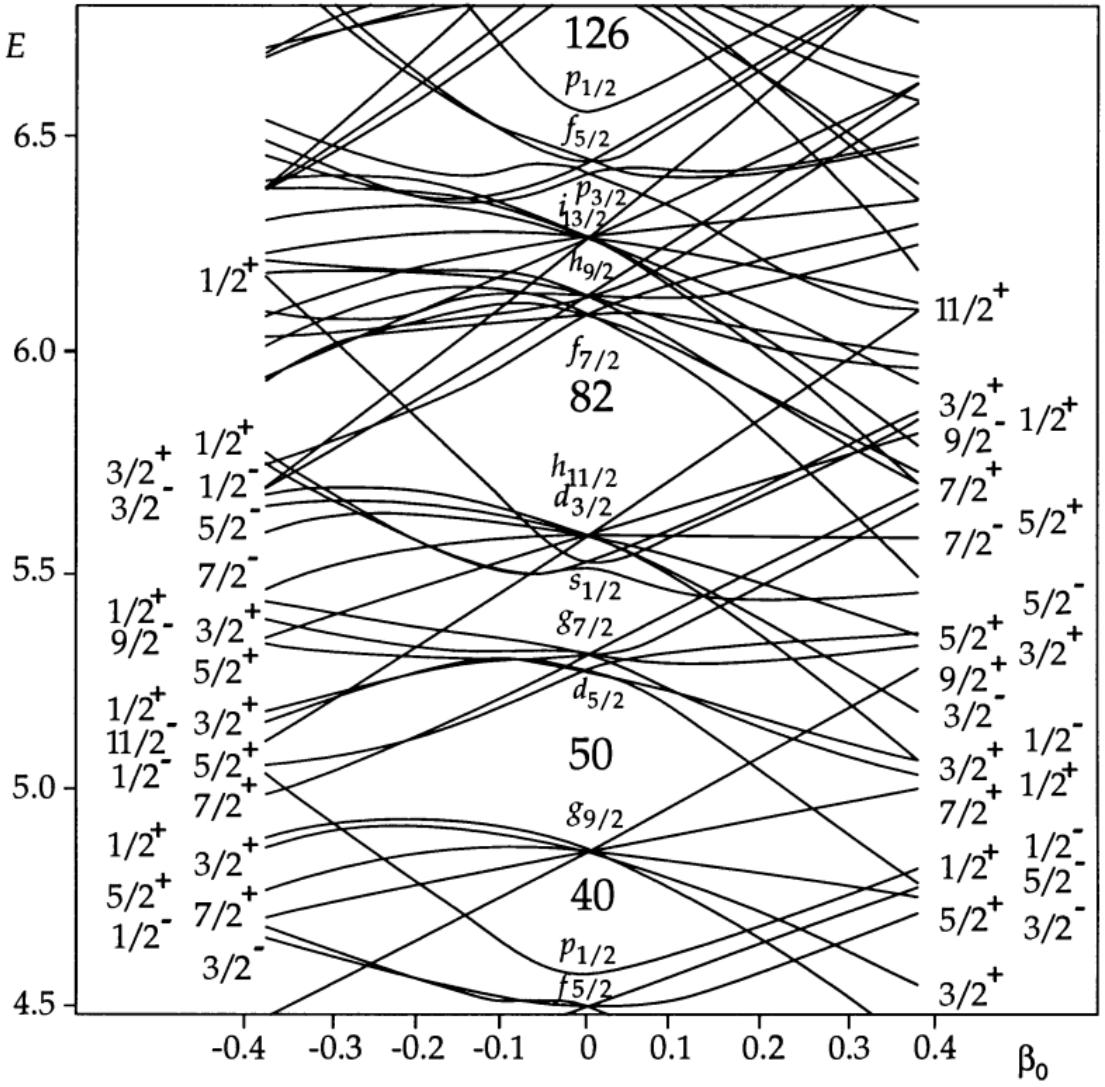


- Our normal Shell Model orbitals are split into has $\frac{2j+1}{2}$ possible projections of K (or Ω)
  - K (or Ω) is defined as the projection of the single particle angular momentum on the nuclear symmetry axis
- For example, the $f_{7/2}$ orbital splits into K between $\frac{1}{2}$ to $\frac{7}{2}$





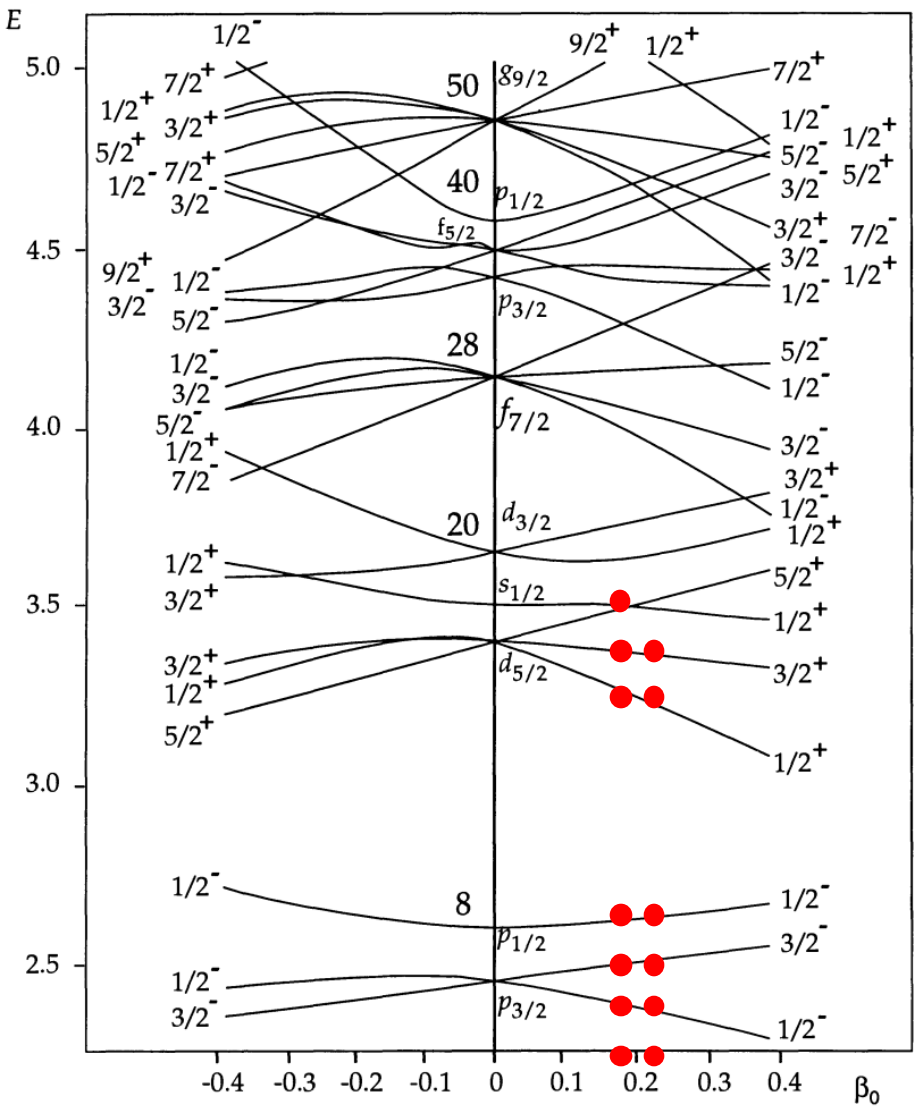




Nuclear Models by Greiner and Maruhn

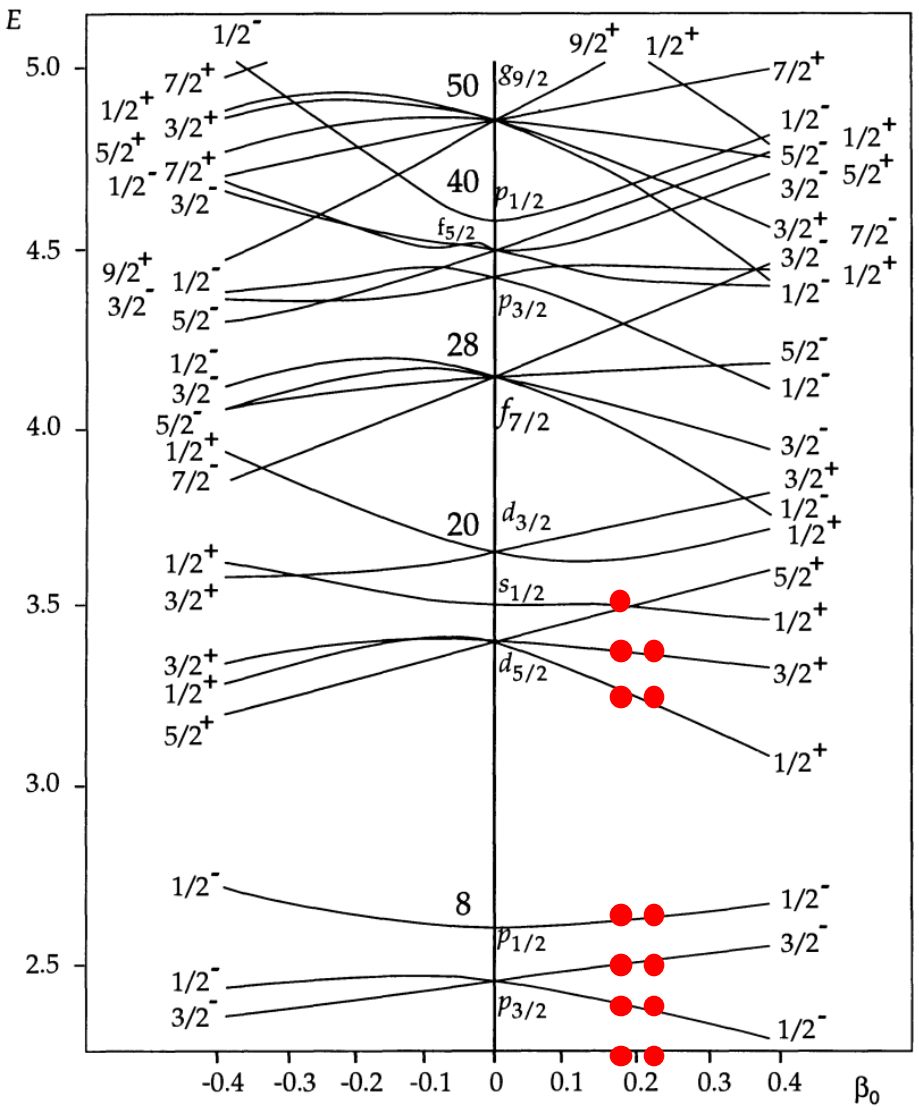
Nilsson Model Example

- Consider ^{25}Al, which we expect to have $\beta_2 \approx 0.2$
- There are 13 protons and 12 neutrons, so the unpaired proton will determine the J^π
- Filling the single particle levels as usual
- So, we predict $J^\pi = \frac{5}{2}^+$
- For the first excited state, it is probable that the proton will move to the $\frac{1}{2}^+$ level above
- This agrees with data



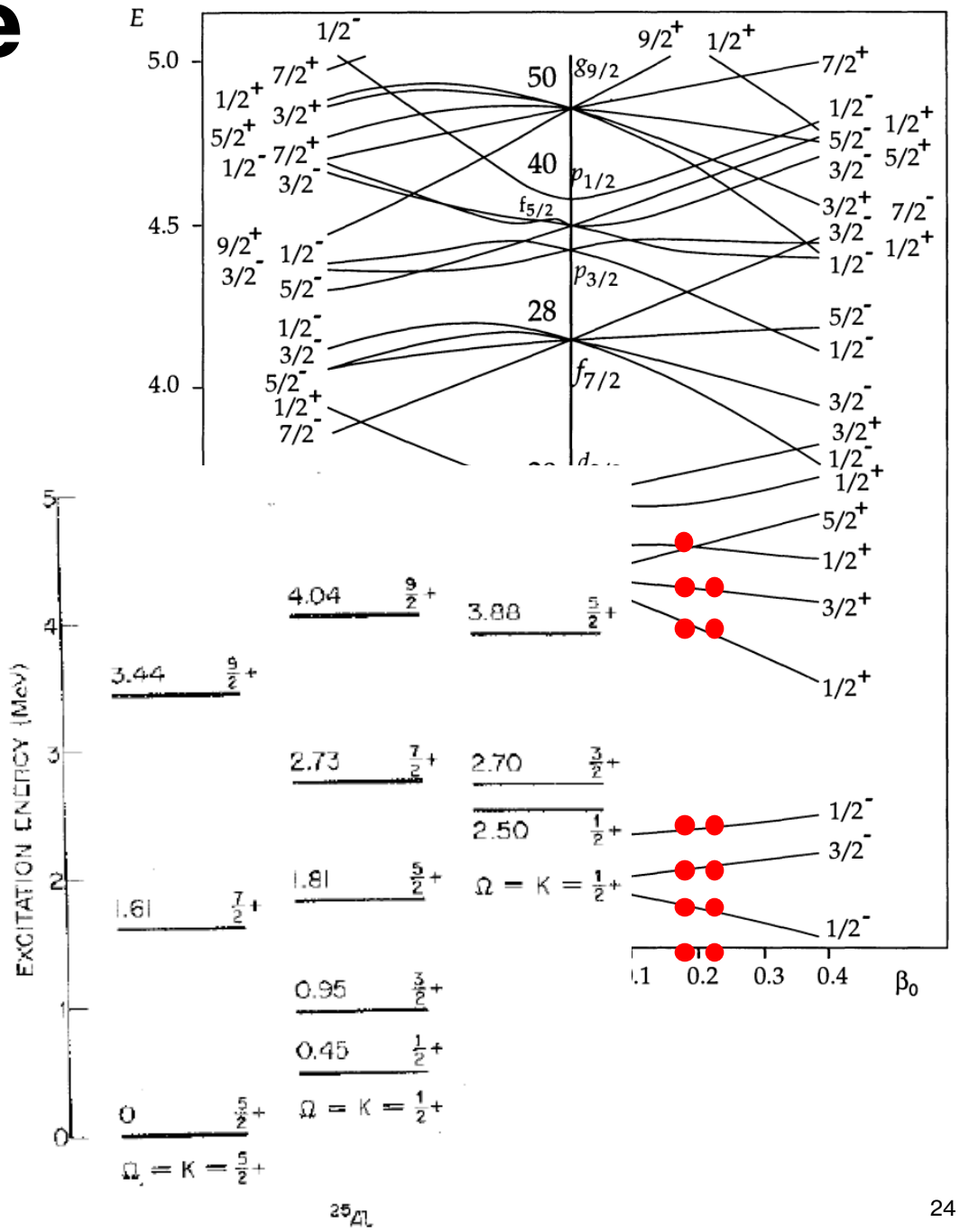
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- This agrees with data
- Since ^{25}Al is deformed, we should see rotational bands with (integer)+j and $\propto j(j+1)$ spacing



Nilsson Model Example

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- For the first excited state, it is probable that the proton will move to the $\frac{1}{2}^+$ level above
- This agrees with data
- Since ^{25}Al is deformed, we should see rotational bands with (integer)+j and $\propto j(j+1)$ spacing

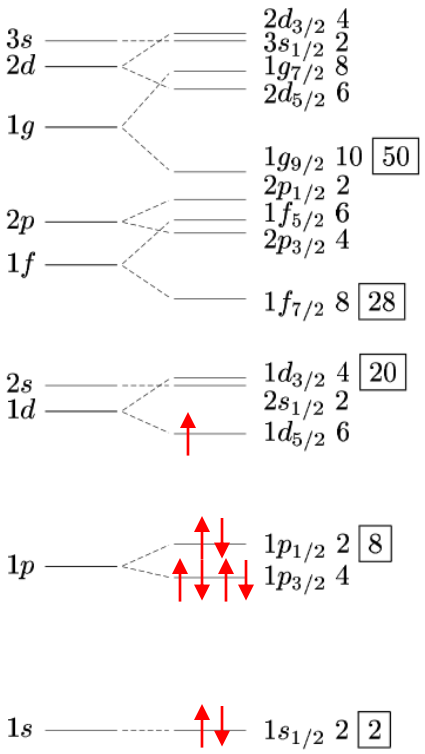


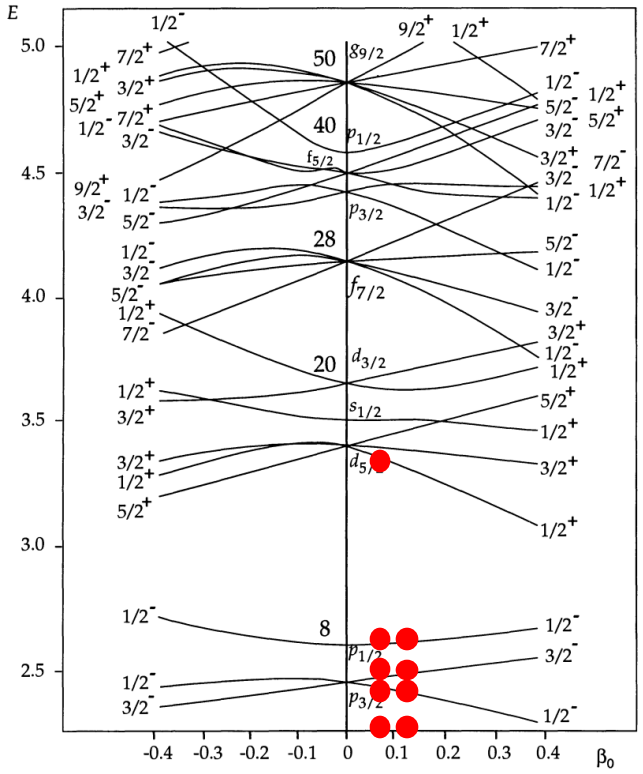
Nilsson Model Group Activity

Compare the non-deformed and deformed shell-model predictions for the ground state of ^{19}F and ^{23}Na where $\beta_2 \approx 0.1$

^{19}F Z=9 N=10

19F		
E(level)	Jn	$T_{1/2}$
0.0	1/2+	STABLE



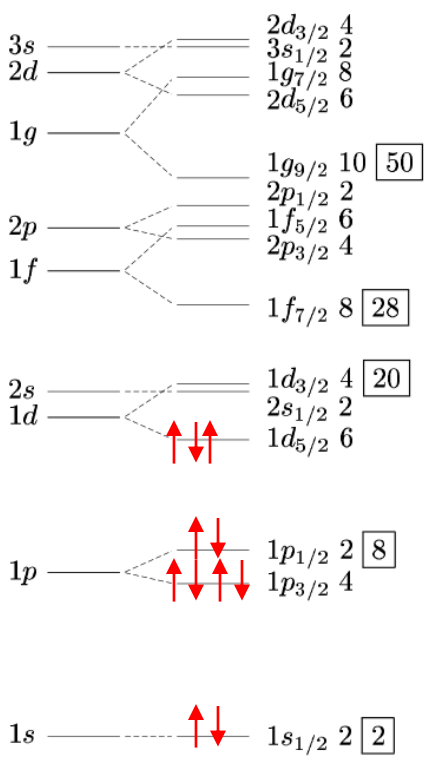


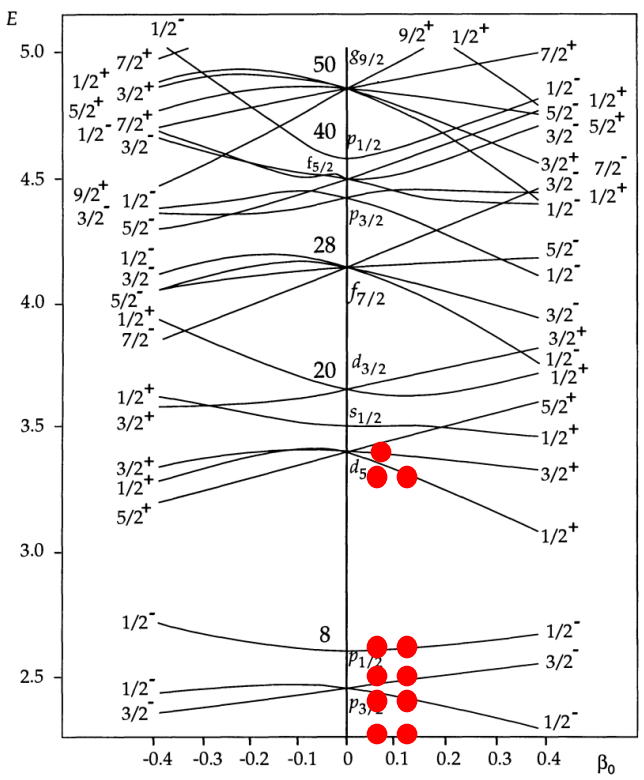
$$J^{\pi} = \frac{5^+}{2}$$

$$J^{\pi} = \frac{1^+}{2}$$

^{23}Na Z=11 N=12

23Na		
E(level)	Jn	$T_{1/2}$
0.0	3/2+	STABLE





$$J^{\pi} = \frac{5^+}{2}$$

$$J^{\pi} = \frac{3^+}{2}$$

Wrapping up

- HW #2 posted last Friday, due Sept 23 at 11:59PM
- Office Hours on Friday at 10AM
    - Edwards 204
- **Next class**: Fermi Gas Model and Radioactive Decay!
- Questions?